Effective Spacetime Geometry from Admissible Non-Injective Projection

Jérôme Beau
Independent Researcher
jerome.beau@cosmochrony.org

2026

Abstract

Continuing from the gauge structure established in the companion paper [8], we derive effective spacetime geometry from the admissibility framework of [1] without introducing it as a postulate. In the continuum limit $q \to \infty$ of [7], the discrete admissibility filter Π_n converges to an effective continuum operator $\mathcal{L}_\Pi$ whose principal symbol defines an effective metric tensor $g^{\mu\nu}(x) \propto A^{\mu\nu}(x)$. The bounded Born–Infeld admissibility constraint (A3–A4) restricts the admissible forms of $A^{\mu\nu}(x)$: in homogeneous regimes, hyperbolicity and spectral stability uniquely select Minkowski geometry; in the presence of a localised stationary obstruction, flux conservation forces the Schwarzschild geometry as the unique stationary exterior solution. The Einstein equations emerge as consistency conditions of emergent geometry, with the Einstein–Hilbert action derived rigorously in [9]. No dynamical substrate is required at any step.

1 Introduction

Position in the programme

The companion paper [8] derives gauge charges, the gauge group, and Yang–Mills dynamics from the admissibility structure of non-injective projection. The present paper addresses the complementary macroscopic question: what effective geometry is selected when the admissibility constraints of [1] are imposed on a spatially varying projection? The answer is Lorentzian geometry with Schwarzschild as the exterior stationary solution (§3).

What is and is not derived in this paper

This paper derives the *structure* of effective spacetime geometry from the admissibility framework. It does not derive the precise numerical value of Newton's constant or quantitative cosmological parameters. The rigorous

Mosco convergence establishing the continuum limit is established conditionally in [7]; the present paper uses it as a working hypothesis throughout.

Organisation

Section 2 establishes the continuum limit of the discrete projection Π_n and the emergence of an effective differential operator. Section 3 derives the effective metric from the principal symbol of $\mathcal{L}_\Pi$ and establishes the emergence of Minkowski and Schwarzschild geometries. Section 4 identifies the open directions.

2 Continuum Limit of the Admissibility Projection

2.1 From discrete Π_n to a continuum operator

The discrete projection Π_n acts on the orbit span $\Omega_n^{(c)} \subset \mathbb{C}^q$, filtering admissible directions via the Born–Infeld bound $A_n \leq c_\chi/\sqrt{\lambda_n}$ (Foundation M, Axiom A3). As established in [1] §3.5, the large-q limit sends:

$$
\begin{aligned}
G_q &\longrightarrow \mathrm{Heis}_3(\mathbb{R}), && (1)\\
\mathbb{C}^q &\longrightarrow L^2(\mathbb{R}), && (2)\\
\text{BFS shells } S_n &\longrightarrow \text{continuous foliation}, && (3)\\
\Pi_n \text{ (discrete)} &\longrightarrow \mathcal{L}_\Pi \text{ (continuum operator)}. && (4)
\end{aligned}
$$

The continuum limit of relational Laplacians on graphs of the type $\mathrm{Cay}(G_q, S_q)$ is governed by standard spectral convergence results [5]: under mild regularity conditions, the graph Laplacian converges in the strong-resolvent sense to a second-order differential operator

$$
\mathcal{L}_\Pi \;=\; \nabla_\mu\big(A^{\mu\nu}(x)\,\nabla_\nu\big), \tag{5}
$$

where $A^{\mu\nu}(x)$ is a symmetric tensor encoding the local connectivity and stiffness of the admissible projection.

Remark 2.1 (Status of the continuum limit)**.** *The convergence $\Pi_n \to \mathcal{L}_\Pi$ is established qualitatively in [1] and rigorously for the Heisenberg Cayley graph $\mathrm{Cay}(G_q, S_q)$ conditionally in [7] (Theorem 7.3, under Hypotheses H1, H-w, H-E1, and numerically confirmed Conjecture C). The present section uses this result as a working hypothesis. A fully rigorous proof closing the remaining hypotheses H1 and H2 is identified as an open direction in Section 4.*

2.2 The operator-metric correspondence

In admissible continuum regimes, the operator $\mathcal{L}_\Pi$ has a well-defined principal symbol:

$$\sigma_2(\mathcal{L}_\Pi)(x,k) \;=\; A^{\mu\nu}(x)\,k_\mu k_\nu. \tag{6}$$

This symbol defines an effective metric tensor [5]:

$$g^{\mu\nu}(x) \;\propto\; A^{\mu\nu}(x). \tag{7}$$

The metric is not an independent dynamical variable: it is a derived object encoding how admissible modes propagate in the continuum limit. This is the operator-metric correspondence on which all subsequent geometric results rest.

3 Effective Spacetime Geometry

3.1 Metric from the principal symbol

The operator-metric correspondence (7) gives the effective metric as $g^{\mu\nu}(x) \propto A^{\mu\nu}(x)$. This metric is not postulated: it encodes how admissible modes propagate in the continuum limit of the projection.

3.2 Signature selection from admissibility

Theorem 3.1 (Lorentzian signature from spectral admissibility, after [5])**.** *In homogeneous and quasi-isotropic admissible regimes, the Born–Infeld admissibility constraint (A3–A4 of [1]) forces the effective metric to have Lorentzian signature* $(-,+,+,+)$.

Proof. The bounded Born–Infeld flux constraint excludes purely diffusive (parabolic) effective descriptions, which would imply instantaneous propagation of perturbations and contradict the finite capacity bound of A4. A Riemannian signature $(+,+,+,+)$ produces an elliptic principal symbol, which is also excluded. An ultra-hyperbolic signature (more than one time direction) fails to admit a globally well-posed initial value problem and is excluded by the spectral stability requirement. The unique remaining option compatible with (i) hyperbolicity of $\sigma_2(\mathcal{L}_\Pi)$, (ii) finite characteristic cone, (iii) the monotonic temporal ordering of Foundation M §4.3, and (iv) the bounded admissibility constraint of A4, is the Lorentzian signature $(-,+,+,+)$. The detailed argument is established in [5] Sections 5–6, which depend only on the bounded-flux constraint and spectral stability — both present in the Foundation M axioms — and not on any dynamical substrate. □

3.3 Schwarzschild geometry from flux conservation

Theorem 3.2 (Schwarzschild from bounded admissibility, after [5])**.** *In the presence of a localised stationary obstruction to the admissibility flow, with spherical symmetry and no additional microscopic scales, the unique exterior stationary solution of the effective geometry induced by $\mathcal{L}_\Pi$ is the Schwarzschild metric*

$$ds^2 \;=\; -\Big(1-\frac{r_s}{r}\Big)\,dt^2 + \Big(1-\frac{r_s}{r}\Big)^{-1}\,dr^2 + r^2 d\Omega^2, \tag{8}$$

where r_s is an integration constant characterising the strength of the obstruction.

Proof. In the exterior region, the homogeneous equation for the effective operator (5) under spherical symmetry reduces to

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2 A^r(r)\frac{d\Phi}{dr}\right) = 0, \tag{9}$$

expressing conservation of admissible flux through spheres of radius r. The general solution is $\Phi(r) = \Phi_0 - C/r$, giving a $1/r$ decay profile. Via the operator-metric correspondence (7), this profile translates to the metric coefficients $g_{tt} = -f(r)$, $g_{rr} = f(r)^{-1}$ with $f(r) = 1 - r_s/r$. The parameter r_s appears as the integration constant of the flux conservation equation and is identified operationally via its influence on admissible propagation cones. This derivation requires only: (i) the bounded-flux constraint (A3–A4), (ii) local injectivity of the projection in the exterior region, (iii) spherical symmetry and stationarity, (iv) absence of additional microscopic scales. No independent metric field equations or energy-momentum tensor are postulated. The detailed argument is established in [5] Sections 6–8. □

Remark 3.3 (Horizon as projection degeneracy)**.** *At $r = r_s$, the lapse function $f(r_s) = 0$. In the present framework, this corresponds to a degeneracy of the principal symbol $\sigma_2(\mathcal{L}_\Pi)$: the operator loses hyperbolicity and the continuum chart ceases to be projectable. The horizon is therefore not a physical singularity of the underlying admissible structure — it is the breakdown of the effective continuum description. This interpretation is consistent with the non-injectivity result of [4]: at the horizon, distinct microscopic configurations become maximally indistinguishable, maximising the projection entropy S_Π.*

3.4 Einstein equations as consistency conditions

Remark 3.4 (Einstein equations from the spectral entropy functional)**.** *The paper [9] shows rigorously, via heat-kernel methods and zeta regularisation, that the renormalized metric variation of the projective entropy*

functional $S_\Pi[g] = \frac{1}{2}\log\det{}'\,\mathcal{L}_\Pi$ produces, in the infrared-dominant two-derivative sector, a response proportional to the Einstein tensor $G_{\mu\nu}$ via the heat-kernel mechanism of Sakharov-induced gravity. The Einstein equations $G_{\mu\nu} = 8\pi T_{\mu\nu}$ therefore emerge as consistency conditions of emergent geometry — not as microscopic laws of the admissible substrate. In the language of the present paper: they express the balance between the curvature of the projection-induced metric (encoding admissibility constraints) and the energy-momentum content (encoding localised obstructions to the admissibility flow). The white paper [6] provides the broader programme context.

4 Open Directions

Rigorous continuum limit. The convergence $\Pi_n \to \mathcal{L}_\Pi$ for the Heisenberg Cayley graph $\mathrm{Cay}(G_q, S_q)$ in the $q \to \infty$ limit is established conditionally in [7]. Closing the remaining hypotheses H1 and H2 and Conjecture C would consolidate the geometric results of the present paper.

Identification of A from admissibility data. The constant $A > 0$ in the continuum Dirichlet form $\mathcal{E}(f,g) = A\int \partial_\mu f\,\partial^\mu g\,d^4x$ must be expressed in terms of the Born–Infeld saturation constant c_χ and the BFS combinatorial data of G_q (Q5a open problem O5). This identification closes the chain from the spectral data to the effective Newton constant $G_N \sim 16\pi^2\ell_{\mathrm{sp}}^2$ of [9].

Coupling constants. The present paper identifies the admissible geometry. It does not determine the numerical values of G_N or observable cosmological parameters. These are expected to emerge from the quantitative spectral data of the O-series via the transfer chain $c_\chi \to \delta_{\mathrm{pair}} \to \beta^* \approx 0.126$, but the explicit identification is open.

Cosmological implications. In homogeneous regimes, the admissibility flow gives a linear growth of cumulative projected capacity $I(n)$, underlying the emergent Hubble law and effective cosmological expansion [6]. The precise connection between the spectral exponent δ_{pair} and the observable cosmological parameters (H_0, w) is an open quantitative problem.

Acknowledgements

The author acknowledges the use of large language models as a supportive tool for refining language, structure, and internal consistency during the development of this manuscript. All conceptual contributions, theoretical choices, and interpretations remain the sole responsibility of the author.

References

[1] J. Beau, "Admissible Non-Injective Transitions as the Primitive of Physical Description," Cosmochrony companion paper M, Zenodo (2026). doi:10.5281/zenodo.19616329.

[2] J. Beau, "Phase Coherence as an Admissibility Constraint: Structural Derivation from Born–Infeld Fibre Minimality," Cosmochrony companion paper M2, Zenodo (2026). doi:10.5281/zenodo.XXXXXXXX.

[3] J. Beau, "Quantum Structure beyond Spin-$\frac{1}{2}$: Co-Admissible Sectors and the Emergence of SU(2) as a Fixed Point," Cosmochrony companion paper M3, Zenodo (2026). doi:10.5281/zenodo.XXXXXXXX.

[4] J. Beau, "Non-Injectivity as a Structural Necessity of Genuine Emergence," Cosmochrony companion paper L, Zenodo (2026). doi:10.5281/zenodo.19391746.

[5] J. Beau, "Bounded Relational Dynamics and the Structural Necessity of Born–Infeld Geometry," Cosmochrony companion paper C, Zenodo (2026). doi:10.5281/zenodo.18407505.

[6] J. Beau, "Cosmochrony: A Pre-Geometric Relational Framework for Emergent Physics," white paper, Zenodo (2026). doi:10.5281/zenodo.17957509.

[7] J. Beau, "Large-q Limit of the Admissible Fibre: Representational and Operator Convergence," Cosmochrony companion paper Q5a, Zenodo (2026). doi:10.5281/zenodo.XXXXXXXX.

[8] J. Beau, "Gauge Structure from Admissible Non-Injective Projection," Cosmochrony companion paper M4a, Zenodo (2026). doi:10.5281/zenodo.XXXXXXXX.

[9] J. Beau, "Infrared Einstein Response from a Renormalized Spectral Entropy Functional," Cosmochrony companion paper H, Zenodo (2026). doi:10.5281/zenodo.XXXXXXXX.

[10] J. Beau, "Weil Representation Spectral Capacity on Heisenberg Cayley Graphs," O12 spectral admissibility sub-programme paper, Zenodo (2026). doi:10.5281/zenodo.19243611.

[11] J. Beau, "Three Stable Directions from Quaternionic Minimality," O23 spectral admissibility sub-programme paper, Zenodo (2026). doi:10.5281/zenodo.19375136.

[12] J. Beau, "Vertical Non-Injectivity and the Stability of the Observable Rank," O24 spectral admissibility sub-programme paper, Zenodo (2026). doi:10.5281/zenodo.19386581.